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Spatial dependence and heterogeneity in landslide susceptibility: A comparative analysis using Gaussian-Based models

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Editor-in-Chief

Computational Statistics and Data Analysis

*Subject: Submission of manuscript titled "**Spatial dependence and heterogeneity in landslide susceptibility: A comparative analysis using Gaussian-Based models**"*

Dear Editor-in-Chief,

Please consider the enclosed manuscript, "Spatial regression models to estimate landslide susceptibility," for publication in *CSDA*.

Landslide susceptibility assessment increasingly relies on data-driven statistical models. However, a critical oversight in many current applications is that these methods are founded on statistical assumptions of spatial independence and homogeneity. This approach neglects the fact that landslide susceptibility analysis is inherently constructed from data with strong spatial patterns, such as clustering and autocorrelation. As we detail in our study, ignoring this fundamental spatial structure violates core regression assumptions, which can lead to biased parameter estimates, misleading significance levels, and ultimately, reduced or overestimated predictive capacity.

In our manuscript, we directly address this methodological gap by proposing and systematically evaluating a suite of spatial regression models grounded in spatial statistics. By analyzing a comprehensive inventory of 13,777 landslides across 533 catchments in the Colombian Andes, we demonstrate how these models explicitly incorporate spatial effects to more accurately reflect the underlying geomorphic reality.

Our findings show that the application of spatial models can increase predictive accuracy by up to 35% compared to standard non-spatial regression. More importantly, our work champions a move away from 'black box' approaches that, despite reporting high accuracy, often fail to improve the physical understanding of the problem. The spatial models presented here not only deliver superior performance but also enhance the interpretability of the results, allowing for a clearer comprehension of the geographic processes and patterns that control landslide occurrence.

We believe this work provides a robust framework for improving landslide susceptibility assessments and will be of great interest to the readers of *CSDA* due to its focus on methodological rigor and its direct implications for natural hazard analysis.

We confirm that this manuscript is original, has not been published elsewhere, and is not currently under consideration by another journal. All authors have approved the manuscript and agree with its submission to *CSDA*.

Thank you for your time and consideration. We look forward to hearing from you.

Sincerely,

Dr. Edier Aristizábal

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Spatial dependence and heterogeneity in landslide susceptibility: A comparative analysis using Gaussian-Based models

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Abstract

Landslide susceptibility assessment refers to a collection of methods aimed at predicting where landslides are likely to occur. Yet, the data used in these models are mostly prone to spatial heterogeneity and dependence, which, if ignored, can lead to biases and inaccurate estimates. In this study, we go beyond the traditional landslide susceptibility framework and use spatial regression methods to estimate landslide frequency in 533 catchments in the Colombian Andes, based on predictors of rainfall, morphometry, geology, and land cover. Moran scatter plots and Lagrange Multiplier tests show that these predictors are spatially autocorrelated and clustered. While landslide occurrence is inherently discrete, we systematically compare an Ordinary Least Squares (OLS) regression against several Gaussian-based spatial models to assess their ability to handle spatial structures in the data. We employ Geographically Weighted Regression (GWR) and Spatial Regimes to address spatial heterogeneity, and Simultaneous Autoregressive (SAR) models to account for spatial dependence. Our results show that accounting for these spatial properties

significantly improves model fit and provides deeper geomorphic insights. GWR and Spatial Regimes reveal distinct regions where relief and elevation exert varying influences, while the superior performance of the Spatial Error Model highlights the impact of spatially autocorrelated, unobserved factors. We conclude that, despite the limitations of a Gaussian framework for count data, integrating spatial dependence and heterogeneity is essential for generating more statistically sound and geographically realistic landslide susceptibility models.

Keywords: landslide; susceptibility; spatial dependency; spatial heterogeneity, Gaussian models, Colombia.

1. Introduction

Landslide susceptibility studies rely on a suite of methods that estimate the likelihood that a slope failure occurs in a particular part of a landscape (Brabb, 1984; Corominas et al., 2014; Fell et al., 2008). Such estimates hinge on statistical models that use a combination of selected geological or environmental predictor variables. The local values of these are routinely obtained for discrete terrain mapping units, which may include regular and arbitrary areas such as grid cells (Ba et al., 2018); irregular but geomorphologically meaningful units such as slope units (Alvioli et al., 2016; Bai et al., 2015); or spatial units that do not necessarily reflect natural boundaries, such as municipality boundaries (Reichenbach et al., 2018).

The response variable in these models is the likelihood of landslide occurrence, which can be expressed in several ways depending on the modeling approach. In most data-driven models, the response variable used for model training is derived from a landslide inventory and is binary, indicating presence or absence of mapped landslides within each mapping unit. Logistic regression models, for instance, predict the probability of classifying a given unit as part of a previously mapped landslide. Other approaches, such as Weights-of-Evidence models (Bonham-Carter et al., 1989), have a continuous response variable that reflects the relative contribution of each predictor.

48 In physically-based models, a training dataset is not required; instead, the response variable is often
49 a continuous and strictly positive value such as the Factor of Safety (Durmaz et al., 2023), which
50 quantifies slope stability. Other ways to express landslide susceptibility include discrete count
51 response variables, such as the number of landslides per unit area modeled using Poisson or
52 negative binomial distributions, or proxies such as landslide density, total affected area, or the
53 proportion of terrain impacted (Reichenbach et al., 2018; Fell et al., 2008). These alternative
54 representations allow for greater flexibility in statistical modeling, enabling the use of different
55 distributions and offering complementary information on landslide hazard in terms of intensity or
56 frequency.

57 Although landslide susceptibility estimation is a spatial modeling problem, most studies assumed
58 that predictors are mutually independent and spatially homogeneous (stationary) (Xiao et al., 2021).
59 This assumption is in conflict with the geospatial nature of the data. Landslide-controlling factors,
60 and the resulting landslide inventories, have some spatial dependence, where nearby landslides are
61 more similar than distant ones, and spatial heterogeneity, where predictors vary across space (Fang
62 et al., 2024; Smith et al., 2023; Liu et al., 2024; Zhang et al., 2023; Quevedo et al., 2022; Lombardo
63 et al., 2020, Anselin, 1988). This spatial structure violates basic assumptions of classical regression
64 models, such as the independence and constant variance of residuals, and negligible
65 multicollinearity among predictors (Cressie, 2015; Ripley, 1988; Anselin, 1988). Violating these
66 assumptions means that some degree of spatial structure remains in the residuals of these models
67 such that part of the variance and its source can be misestimated (Opitz et al., 2022; Rey et al.,
68 2023). Spatial heterogeneity can lower the validity of model estimates, bias parameter estimates,
69 create misleading significance levels, and poor forecasts (Li et al., 2022). Similarly, spatial
70 dependence may produce misleading significance levels of T-tests, and inaccurate R^2 -values due to
71 biased estimates of the error variance (Anselin, 1990).

72 Despite these caveats, it remains a widespread practice to ignore spatial structure when estimating
73 landslide susceptibility (Fotheringham et al., 2000; Schabenberger & Pierce, 2001). Meanwhile,
74 spatial models have become popular for addressing dependency and heterogeneity in fields such as
75 ecology and disease mapping (Bivand et al., 2021), under the term Bayesian hierarchical regression
76 models (Haining & Li, 2020) and spatial econometrics (Anselin, 1988; Arbia, 2016). These
77 techniques capture spatial relationships and local variations that traditional models fail to account
78 for, but have hardly been explored in landslide susceptibility studies (Di Napoli et al., 2023; Fang et
79 al., 2024; Liu et al., 2024), partly because their complexity (Bryce et al., 2024).

80 Spatial modeling generally follows one of two main strategies. One is to treat data as if they were
81 obtained from a continuous field, and appropriate models include Gaussian Processes (Rue & Held,
82 2005), for example kriging (Cressie, 2015). The spatial structure of the data is encoded by a
83 covariance matrix based on the pairwise distances between the observations. Thus, Gaussian
84 Processes are computationally demanding for large datasets with commensurately large covariance
85 matrices (Banerjee et al., 2008), likely limiting this method for landslide susceptibility (Bryce et al.,
86 2022).

87 The second strategy considers data in discrete spatial units and includes two main families of
88 models: autoregressive models and spatially varying coefficient (SVC) models. Autoregressive
89 models (Anselin, 1996; Yadav et al., 2023) introduce spatial structure through a covariance function
90 that links each observation with its neighbors, rather than with all other observations as in Gaussian
91 Process models (Fischer & Wang, 2011; Fotheringham et al., 2000). This approach allows for
92 several ways of defining neighborhood relationships among discrete data (Getis, 2009) and is
93 computationally efficient, straightforward to implement, and easily interpretable in relation to
94 nearby units. In contrast, SVC models allow regression coefficients to vary across space, either at
95 the level of each individual unit or over spatially homogeneous regions.

96 This study focuses on explicitly integrating and evaluating a Gaussian-based spatial regression
97 framework to model landslide susceptibility in the Colombian Andes. First, we address a common
98 methodological pitfall in landslide susceptibility by systematically quantifying the extent of
99 misestimation that occurs when using a non-spatial Ordinary Least Squares (OLS) model. While
100 most studies favor logistic regression or machine learning classifiers, we compare OLS against a
101 suite of spatial models specifically designed to manage spatial heterogeneity and dependence,
102 highlighting the critical information lost when spatial structures are neglected. Second, we use this
103 framework as a diagnostic tool to investigate how accounting for spatial structure provides a more
104 nuanced, geomorphologically-interpretable understanding of landslide controls. We argue that these
105 models are not just statistical refinements, but essential tools for revealing how the influence of
106 predictors such as relief or elevation varies across the landscape, an insight obscured by traditional,
107 spatially-static models.

108 **2. Spatial regression models for discrete data**

109 Spatial models consider two types of spatial structure in data, i.e. heterogeneity and dependence
110 (Anselin, 1988). These terms are also known as non-stationarity and autocorrelation in time series
111 analysis, respectively. Unlike time series that are one-dimensional, spatial data can involve more
112 dimensions. Figure 1 summarizes various spatial regression models and their relationships. These
113 include models designed for continuous spatial fields, such as Gaussian Processes (GP), and models
114 for spatially discrete data, which can be further categorized into autoregressive (AR) models that
115 address spatial dependence, and spatially varying coefficient (SVC) models that capture spatial
116 heterogeneity.

117 **2.1 Spatial heterogeneity**

118 Spatial heterogeneity refers to domain effects on the sample units, where the local mean as a first-
119 order statistical moment varies from place to place (Zhang et al., 2023; Wang et al., 2022). Spatial

heterogeneity in landslide susceptibility invites using two types of SVC models, i.e. Geographically Weighted Regression and Spatial Regimes.

The most common SVC model is Geographically Weighted Regression (GWR) (Brunsdon et al., 1996; Fotheringham et al., 2009). In GWR models, the relationship between predictors and the response varies continuously across geographic space using a local weighting matrix (Peng & Inoue, 2024). GWR enables us to examine how geomorphic drivers of landsliding vary spatially, testing the hypothesis that the explanatory power of each predictor is modulated by local geographic settings rather than being a universal constant. For instance, the effect of slope on landslide occurrence might be very strong in coherent bedrock but weak in unconsolidated sediments. GWR produces a continuous surface of coefficient estimates of where and how these relationships change. This method captures spatial variation by assigning greater weight to nearby observations using a kernel function, such as Gaussian, exponential, or bi-square (Fotheringham et al., 2000; Opitz et al., 2022). This kernel function needs a specified bandwidth that defines the neighborhood of influence around each point. The bandwidth can be either fixed in terms of a geographic distance, or adaptive by adjusting to a set number of neighboring observations (Brunsdon et al., 1996). The general formulation of a GWR model for a given location i is:

$$y_i = \beta_{i0} + \sum_{k=1}^k \beta_{ik} x_{ik} + \varepsilon_i$$

where y_i is the value of the response variable; β_{i0} is the intercept; β_{ik} is the k th coefficient, x_{ik} is the k th predictor variable; k is the number of predictors, and ε_i is the random error term.

Spatial Regime (SR) models belong to the family of multilevel models, where observations are assigned to distinct groups or regimes that represent conceptually defined regions (Anselin, 1996). These regimes function as levels in the multilevel framework, allowing for the examination of structural differences between regions. In SR models, regression coefficients are estimated for each

143 regime, capturing how relationships between predictors and the dependent variable may differ
144 across these predefined spatial zones. The spatial pattern in these models is determined by the
145 geographical or contextual delineation of regimes, which can be based on physical boundaries (e.g.,
146 basins or spatial clusters), land use classes, or other spatial criteria (Piras & Sarrias, 2023). Rather
147 than modeling continuous spatial variation, SR models focus on within-regime variations while
148 allowing comparisons of parameter differences across regimes. This approach retains the multilevel
149 character of the model, as it accounts for both within-regime effects (individual-level variation) and
150 between-regime differences (group-level variation). SR models can also include spatially fixed
151 effects, where each regime has its own set of coefficients.

152 **2.2 Spatial dependency**

153 Spatial dependence means that adjacent observations are more similar or correlated than those
154 farther apart (Fletcher & Fortin, 2018), estimated by the covariance as a second-order moment
155 (Wickham, 2008). This spatial dependence is modeled by incorporating a spatial lag as a predictor,
156 which represents a weighted average of neighboring values. The spatial lag is computed using a
157 neighborhood connectivity matrix $\mathbf{W}$ that defines the spatial relationships between units (Anselin,
158 1996; Rey et al., 2023).

159 Both Conditional Autoregressive (CAR) and Simultaneous Autoregressive (SAR) models can
160 handle spatial dependency in discrete areal data by relying on neighborhood information (Wall,
161 2004; Jaya & Chadidjah, 2021; Cressie, 2015). SAR models explain relationships among response
162 variables at all locations simultaneously (Ver Hoef et al., 2018; Jaya & Chadidjah, 2021), whereas
163 CAR models estimate local response conditional on values at neighboring locations (Anselin,
164 1988). CAR models are defined within the framework of Bayesian Multilevel models, where spatial
165 dependence is introduced as a structured random effect through a Markov Random Field.

166 In SAR models, the treatment of spatial dependence across all locations gives rise to what is known
 167 as a spillover effect—that is, a change in a predictor or in the response variable at one location can
 168 propagate and indirectly influence values at other locations, including those not directly connected
 169 through the neighborhood matrix (LeSage, 2014). The spillover effect quantifies how a change in a
 170 predictor at one location (i) can influence the response variable at other locations (j) (LeSage, 2014;
 171 Bivand, 2022). While this effect is mathematically well-defined within the spatial weighting matrix,
 172 its physical interpretation in a landslide context must be handled with caution. This property makes
 173 SAR models suited for capturing spatial diffusion processes and global interaction effects.

174 Neighborhood connectivity matrices can be obtained in several ways, i.e. by (i) p -order binary
 175 contiguity matrices that consider p -order neighbors (where p is a positive integer representing first-
 176 order neighbors $n=1$, second-order neighbors $n=2$, and so on, up to $p=n$ levels); (ii) inverse distance
 177 matrices or exponential distance decay matrices; or (iii) q -nearest neighbor matrices, where q is a
 178 positive integer representing the number of neighbors considered (Getis & Aldstadt, 2004; Getis,
 179 2009; Stakhovych & Bijmolt, 2009).

180 In SAR models, spatial lag can be introduced in different model terms, leading to three main types
 181 of spatial dependency: (i) exogenous autocorrelation, where the spatial lag is applied to the
 182 predictor variables ($\mathbf{WX}$); (ii) endogenous autocorrelation, where the spatial lag affects the response
 183 variable ($\mathbf{Wy}$); and (iii) autocorrelation in the error term, where spatial structure is in the residuals
 184 ($\mathbf{W}\mu$), leaving the predictors uncorrelated with the composite error (Anselin, 1988). The General
 185 Nesting Spatial (GNS) model—also known as SARAR(1,1) or the Cliff-Ord spatial model—
 186 encompasses all three forms of spatial dependence (LeSage & Pace, 2009; Kelejian & Prucha,
 187 2007; Cliff & Ord, 1981):

$$188 \quad y = X\beta + WX + \rho Wy + \lambda W\mu + \epsilon$$

189 where the positive $n \times n$ spatial weight matrix $\mathbf{W}$ describes the structure of dependence between n
190 observations in a $n \times 1$ vector y , in which each element y_i represents the response variable in spatial
191 unit i ; β is a $k \times 1$ vector of k coefficients for the predictors in design matrix $\mathbf{X}$ with $n \times k$ dimensions;
192 ρ is the coefficient of the spatially lagged dependent variable; λ is the coefficient in a spatial
193 autoregressive error; μ and ε are the spatial and local error terms of the model. Here, μ follows a
194 first-order spatial autoregressive process, whereas ε is a $n \times 1$ vector of independently and identically
195 distributed errors with zero mean and variance σ^2 . The term $\rho\mathbf{W}$ captures endogenous interaction
196 effects of the response variable, while $\mathbf{W}\mathbf{X}$ models the exogenous interaction effects among the
197 predictors; the term $\lambda\mathbf{W}$ deals with the interaction effects among the residuals.

198 Among the simplified variants of the SAR model, the Spatially-lagged y Model (SLy), where $\lambda=0$
199 and $\mathbf{W}\mathbf{X}=0$, thus only featuring the spatially lagged dependent variable, $\rho\mathbf{W}y$. In the case that $\rho=0$
200 and $\mathbf{W}\mathbf{X}=0$, the Spatial Error Model (SEM) retains the error terms $\lambda\mathbf{W}\mu$, while the Spatial Lagged X
201 Model (SLX) comprises the spatial lags of the predictors, $\mathbf{W}\mathbf{X}$, when $\lambda=0$ and $\rho=0$ (Elhorst, 2013).
202 Further variants include the Kelejan-Prucha or Spatial Autoregressive Combined Model (SAC) that
203 includes autocorrelation in $\rho\mathbf{W}y$ and the $\lambda\mathbf{W}\mu$; the Spatial Durbin Model (SDM) combines an
204 autoregressive dependent variable $\rho\mathbf{W}y$ and spatially lagged predictors $\mathbf{W}\mathbf{X}$; and the Spatial Durbin
205 Error Model (SDEM) has a spatial error term $\lambda\mathbf{W}\mu$ combined with spatially lagged predictors $\mathbf{W}\mathbf{X}$
206 (Elhorst, 2022; Anselin, 2022); (Fig. 1).

207 In models with endogenous dependence (e.g., SLy, SDM), the spillover effect is global: a change in
208 one catchment propagates through the system-wide connectivity defined by the spatial weights
209 matrix ($\mathbf{W}$). Due to the simultaneous feedback nature of the autoregressive term (ρ), an impact in
210 one unit simultaneously feedback its neighbors, then to the neighbors of those neighbors, eventually
211 influencing the entire study area. This global spillover implies a form of systemic propagation that
212 is physically difficult to justify for landslides, as they are discrete slope failures rather than a
213 contagious process. Instead, this mathematical dependence suggests that the spatial structure is

likely driven by the predictors; the environmental drivers within each mapping unit exhibit strong spatial autocorrelation, which in turn conditions the distribution of landslide counts. Therefore, the model captures the spatial organization of the susceptibility factors rather than a direct physical interaction between individual landslide events. Conversely, in models with only exogenous dependence (e.g., SLX, SDEM), the spillover is restricted to the immediate spatial neighborhood defined by $\mathbf{W}$. This framework is geomorphologically more plausible as it represents localized interactions; for example, it accounts for how intense rainfall or specific geological conditions in an adjacent catchment—rather than across the entire study area—may influence the landslide count within a given unit. In this sense, the model captures the 'neighborhood effect' of environmental drivers, reflecting the spatial scales at which triggering factors typically operate. The SEM model, by contrast, confines all spatial dependence to the error term and assumes no spillover in the response variable. Therefore, selecting among these models is not just a statistical choice; it is a decision about the assumed spatial nature of the underlying environmental and landscape processes, whether they are localized, systemic, or driven by latent, i.e. unobserved, factors (Elhorst, 2013).

2.3 Spatial Model Tests

The first step in exploring the spatial structure of predictors is to test for autocorrelation. The Moran scatter plot and its associated Moran's I statistic, and the Lagrange Multiplier (LM) test for omitted spatial effects (Anselin, 1996). The scatter plot relates each standardised observation to the spatially lagged mean of its neighbours (Anselin, 1988); the slope of a least-squares fit is Moran's I ,

$$I = n \frac{\sum_{i=1}^n \sum_{j=1}^n w_{ij} (y_i - \bar{y})(y_j - \bar{y})}{(\sum_{i=1}^n \sum_{j=1}^n w_{ij})(\sum_{i=1}^n (y_i - \bar{y})^2)}$$

where n is the number of observations, y_i is the value of the response variable at location i , $\bar{y}$ is its mean, and w_{ij} are elements of the row-standardised spatial-weight matrix $\mathbf{W}$. Moran's I ranges from -1 (perfect dispersion) to +1 (perfect clustering), with zero indicating spatial randomness. Because global measures can mask local spatial patterns, we also compute Local Indicators of Spatial

238 Autocorrelation (LISA) (Anselin, 1996). LISA decomposes Moran's I into location-specific
239 contributions and obtains pseudo- p values via permutation, thus pinpointing statistically significant
240 "hot spots", "cold spots", and spatial outliers.

241 The LM test is used to decide on an appropriate spatial regression model and to estimate spatial
242 autocorrelation in both the response variable and its residuals. The test compares a ordinary least
243 squares model with unrestricted models that have spatial terms. The spatial term can either be a
244 spatially lagged dependent variable or a spatially lagged error term. The LM statistic is calculated as
245 the difference between the log-likelihoods of the restricted and unrestricted models, following a χ^2
246 distribution with one degree of freedom under the null hypothesis of no spatial dependence. If the
247 test statistic exceeds the critical value, the null hypothesis of spatial independence is rejected.

248 **3. Data**

249 **3.1 Study area**

250 Our study area is located north of 5°N and encompasses the Western and Central Cordilleras of the
251 Colombian Andes, separated by the Cauca canyon. The region is bound by the Atrato River to the
252 west and the Magdalena River to the east. Covering 50,000 km², the area has 533 catchments of
253 medium order within the Atrato (25% of catchments), Cauca (50%), and Magdalena basins (25%).
254 Roughly 73% of these catchments drain areas less than 100 km²; the median area is 48 km².

255 **3.2 Response variable**

256 We used a catalogue of 13,777 shallow and deep-seated landslides that occurred between 1970 and
257 2023 (Aristizábal & Korup, 2025) (Fig. 2), detected manually using high-resolution (<1 m) true-
258 color optical images from Google Earth™. While this inventory provides a comprehensive regional
259 dataset, we acknowledge several uncertainties due to its compilation from optical imagery (Google
260 Earth™). The manual detection process is subject to several potential sources of spatial bias, such
261 as variable image resolution and cloud cover across the study area, which may lead to under-

reporting in certain regions. This inventory is thus considered a representative, albeit incomplete, sample of landslide occurrence.

We use landslide abundance—the number of landslides per spatial unit—as the response variable. Instead of merely classifying a catchment as susceptible or not, modeling landslide abundance allows us to differentiate between areas based on their relative levels of observed landslide occurrence, offering a more comprehensive and informative view of landslide susceptibility. In our case, the empirical distribution of landslide abundance in the Colombian Andes is characterized by a high frequency of zero counts and a strong positive skew. Our modeling approach is based on regression techniques that assume a Gaussian distribution, which is not applicable to count data. To adapt our data to this framework, we transformed the response variable using a logarithmic function ($\log(Y + 1)$); adding 1 avoids zero landslide number values, whose logarithm would produce undefined results. This step transforms the skewed, discrete data into an approximately continuous variable, stabilizes the variance, and helps the subsequent model's residuals approach a Gaussian distribution (Figure 3).

3.3 Predictor variables

Initially, we selected nine geomorphological, geological, and climatic candidate predictors to model landslide abundance at the catchment scale, drawing on their reported relevance in earlier studies (Reichenbach et al., 2018; Van Westen et al., 2008). These predictors include: the hypsometric integral (H_i), which reflects the stage of landscape evolution and potential for landsliding; mean elevation (E) and mean slope (S), both of which influence gravitational driving forces; mean local relief (H), indicating terrain ruggedness; lineament density (Ld) representing structural controls such as faults and fractures that can weaken slopes; mean annual rainfall (P), a key factor in weathering processes; and land use cover (Lc) and geology (G), which reflect soil properties, vegetation cover, and material strength. We also included the number of days with daily rainfall

286 exceeding 20 mm (P_{20}) because short-term intense rainfall events are commonly associated with
 287 landslide initiation in the Colombian Andes (Aristizábal & Sánchez, 2020).

288 To align all predictors with the catchment-level response variable, we aggregated their values within
 289 each catchment: continuous variables were averaged, lineament density was calculated, and
 290 categorical variables were represented by their modal class. All terrain parameters were calculated
 291 using the 12.5-m digital elevation model (DEM) from the Advanced Land Observing Satellite-
 292 Phased Array-Type L-Band Synthetic Aperture Radar (ALOS-PALSAR) (Logan & Karr, 2014).
 293 Lineament density (Ld) was estimated using the LINE algorithm from the image processing
 294 software CATALYST, based on SRTM data (Villalta & Smith, 2002), and for eight hillshade
 295 directions at 45° bins of azimuth with a sun elevation angle of 45°. We derived rainfall information
 296 from the Climate Hazard Group InfraRed Precipitation with Station Data (CHIRPS) (Funk et al.,
 297 2015), version 2.0, at 5-km resolution from 1981 to 2023, using Google Earth Engine (GEE). We
 298 generated the land cover map using 10-m Sentinel-2 imagery from the Copernicus Programme. We
 299 selected images from 2018 to 2023 with <10% cloud coverage, removed shadow effects, and
 300 created layers of the Normalized Difference Vegetation Index (NDVI) (Kriegler, 1969), Normalized
 301 Difference Built-up Index (NDBI) (Zha et al., 2003), Modified Normalized Difference Water Index
 302 (MNDWI) (Xu, 2006), and Bare Soil Index (BSI) (Li et al., 2014). We performed land cover
 303 classification using a random forest algorithm implemented with the GEE Classifier package
 304 (Breiman, 2001).

305 To implement SR models, it is necessary to establish spatial regimes or homogeneous regions. We
 306 tested two regimes: the first regime reflects the major drainage basins in our study area, i.e. Atrato,
 307 Cauca, and Magdalena. However, this hydrological segmentation may not necessarily fully explain
 308 the abundance of landslides. Hence, for the second regime we used spatial clustering on
 309 morphometric parameters to identify homogeneous regions that might explain the spatial patterns of
 310 landslide occurrence. We used Agglomerative Clustering using K-nearest neighbors for $K \in \{2, \dots,$

6}, using the Python Scikit-Learn library, and guided by the elbow method and Silhouette coefficient (Pedregosa et al., 2011). The elbow method helps identify the optimal number of clusters by plotting the Within-Cluster Sum of Squares (WCSS) against K to detect an inflection point (Thorndike, 1953). The Silhouette coefficient (Sc) assesses cohesion within, and separation between, clusters, and ranges from -1 (poor clustering) to $+1$ (strong clustering) (Rousseeuw, 1987). For handling spatial heterogeneity, we used the Python MGWR library by Oshan et al. (2019), and the PySAL library by Rey & Anselin (2007). For estimating spatial dependency, we used the R libraries spdep and spatialreg by Bivand (2022). The impacts function from the R package spatialreg helped to estimate direct and indirect impacts of predictors.

3.4 Model performance

To compare the performance of all selected models, we use Akaike's Information Criterion (AIC) along with the adjusted R^2 or the Nagelkerke R^2 . The AIC evaluates model performance by balancing goodness of fit against the number of estimated parameters. The choice between adjusted R^2 and Nagelkerke R^2 depends on the model type. For the non-spatial OLS model, we report the adjusted R^2 to measure the proportion of variance explained by the predictors while accounting for the degrees of freedom. For models estimated via Maximum Likelihood, we use the Nagelkerke pseudo- R^2 (Nagelkerke, 1991), which provides a comparable measure of explanatory power by normalizing the likelihood ratio. These metrics allow us to evaluate the goodness of fit across different estimation frameworks while penalizing the inclusion of unnecessary parameters. In contrast, for models estimated through maximum likelihood estimation (MLE)—such as SAR and SR models—we use the Nagelkerke R^2 , which expresses the improvement of model likelihood relative to a null model, and is scaled between 0 and 1. We include these measures to provide an intuitive understanding of model performance, with AIC remaining the primary criterion for selecting the most parsimonious model.

335 **4. Results**

336 **4.1 Variable selection**

337 To reduce multicollinearity and reduce the dimensionality of the environmental dataset, we first
338 conducted a Principal Component Analysis (PCA) on the set of standardized predictor variables.
339 We retained components with eigenvalues greater than 1 (Kaiser's criterion (Kaiser, 1960)) and
340 used the loading scores to group collinear variables. Variables with high loadings (>0.6) on the
341 same principal component were considered collinear and evaluated for exclusion or aggregation.

342 The first two principal components explained 70% of the total variance—with PC1 accounting for
343 49% and PC2 for 21%. Figure 4 illustrates the PCA results, highlighting four groups of collinear
344 predictors: (i) lineament density (Ld), mean local relief (H) and mean slope (S); (ii) hypsometric
345 integral (H_i) and mean elevation (E); (iii) mean annual rainfall (P) and days with rainfall > 20 mm
346 (P_{20}); and (iv) Area (A). Predictors Lc, G and Ld have coefficient estimates that may hardly differ
347 from zero. Conversely, Ld shows high correlation values with H (0.81) and S (0.91), while H and S
348 correlate (0.96). Similarly, the two rainfall predictors, P and P_{20} , are highly correlated (0.98), and
349 negatively correlated with E (-0.6). Although, H_i shows low correlation values with the other
350 predictors, the coefficients were very low and had no statistical significance.

351 Within each group we selected a single representative predictor while discarding the remaining
352 collinear predictors. Categorical variables (land-use cover Lc, geology G) were excluded from the
353 PCA, and we retained them in their modal form because of their clear interpretative value. This
354 screening step provides a parsimonious, interpretable set of candidate predictors for the subsequent
355 regression analysis.

356 **4.2 Exploratory spatial structure**

357 To define the spatial weight matrix, we used the k -nearest-neighbour (k-NN) approach, which
358 guarantees that every catchment has at least one neighbor and avoids isolated units with no spatial

359 links. Moreover, it allows weighting neighbor influence as a function of distance, assigning greater
360 weight to closer catchments. We evaluated matrices with $2 < k < 9$, and used Moran's I and Local
361 Indicators of Spatial Association (LISA) to identify the value of k that maximized both global and
362 local autocorrelation. Results indicated that $k=5$ provided the highest spatial coherence while
363 keeping $\mathbf{W}$ sparse. Figure 5 shows the resulting spatial connectivity network, which we use for all
364 subsequent analyses.

365 Using the $k=5$ matrix, we computed Moran's I and Local Indicators of Spatial Autocorrelation
366 (LISA) for every morphometric covariate. Slope, elevation, and relief have a strong positive spatial
367 autocorrelation ($I > 0.8$), whereas H_i has a moderate structure. The LISA map for slope (Fig. 6)
368 shows statistically significant hot spots (High–High) and cold spots (Low–Low).

369 To delineate spatial regimes we ran a k-means clustering of the standardised morphometric
370 variables, using the same $k=5$ weight matrix as spatial constraints. The optimal solution, obtained
371 by minimising the within-cluster sum of squares and maximising the silhouette width, has four
372 clusters (A–D; Figure 7). Cluster A corresponds to the low-lying alluvial terrain along the
373 Magdalena and Cauca basins; cluster B represents the high-relief terrain along the Cauca canyon;
374 cluster C is the low alluvial terrain along the Atrato basin, and cluster D comprises the high-
375 elevation, low-relief surfaces along the axis of the central Cordillera. Both regimes (basin and
376 morphometric clusters) are used in the spatial models.

377 **4.3 Standard linear regression model (non-spatial)**

378 Using the reduced predictor set obtained, we fitted several multivariate, non-spatial regression
379 models with OLS estimation. Table 1 lists the parameter estimates for the standard linear regression
380 models based on various combinations of standardized predictors. These non-spatial OLS models
381 serve as the benchmark for gauging the benefits of incorporating spatial effects in more
382 sophisticated models.

383 Table 1: Parameter estimates for non-spatial, standard regression models using standardized
384 predictor variables. Bold values indicate statistically significant parameter estimates at $p < 0.05$, with
385 standard errors in parentheses; model performance is expressed by Adjusted R^2 and Akaike
386 Information Criterion (AIC).

Predictor	OLS1 (all predictors)	OLS2	OLS3	OLS4	OLS5 (selected predictors)
Constant	1.814 (0.074)	1.840 (0.056)	1.862 (0.086)	1.759 (0.076)	1.840 (0.048)
<i>A</i>	0.586 (0.050)	-	-	0.603 (0.051)	0.557 (0.049)
<i>H_i</i>	0.069 (0.058)	0.104 (0.063)	0.035 (0.067)	-	-
<i>Ld</i>	0.179 (0.150)	-0.449 (0.142)	-0.024 (0.106)	-	-
<i>P</i>	-0.077 (0.067)	-0.253 (0.063)	-0.272 (0.066)	-	-
<i>E</i>	0.423 (0.074)	-	-	-	0.514 (0.058)
<i>S</i>	-0.123 (0.298)	1.112 (0.146)	-	0.415 (0.191)	-
<i>H</i>	0.573 (0.204)	-	0.792 (0.103)	0.492 (0.182)	0.546 (0.058)
<i>Lc</i> (grass)	0.202 (0.123)	-	0.197 (0.143)	0.504 (0.119)	-
<i>G</i>	-0.002 (0.171)	-	-0.242	-0.293	-

Predictor	OLS1 (all predictors)	OLS2	OLS3	OLS4	OLS5 (selected predictors)
(sediment)			(0.194)	(0.173)	
G (volcanic)	-0.131 (0.125)	-	-0.189 (0.145)	-0.135 (0.126)	-
Adj. R ²	0.500	0.319	0.318	0.445	0.497
AIC	1611.636	1768.473	1772.166	1662.796	1608.693

387

388 The best OLS model of the ones tested here was that with predictors catchment area *A* and terrain
389 parameters *H* and *E*, and landslide catchment abundance tends to be significantly higher with
390 increasing local relief and higher elevations. This model explains only 50% of the variance in the
391 landslide number per catchment. We examined the model residuals for spatial structure (Figure 8)
392 both globally and within specific spatial regimes. By analyzing the spatial distribution of residuals
393 across basins and morphometric clusters, we sought to identify regions where the model
394 consistently over- or under-predicts landslide abundance. Such spatial variations in residual patterns
395 suggest that the underlying geomorphological or hydrological drivers may operate differently across
396 regimes, indicating that a single global parameters set may not fully capture the local complexities
397 of the landscape.

398 Residuals are higher in catchments in the Cauca and Magdalena basins, indicating poorer model fit
399 in these regions. Similarly, clusters B and D, where landslides are most abundant, show higher
400 residual variances, suggesting that local terrain variations may not be fully captured by the non-
401 spatial model. The Moran's *I* for the global residuals of 0.376 ($p = 0.001$) indicates spatial
402 autocorrelation. Similarly, the Lagrange Multiplier (LM) tests yield significant statistics for both the
403 error term ($RS_{err} = 205.64$) and the spatial lag of the dependent variable ($RS_{lag} = 182.90$). The

404 higher value for the error term suggests that a spatial error model (SEM) is appropriate to address
 405 the residual spatial structure. This suggests that the model is failing to account for spatially-
 406 clustered latent drivers—such as regional tectonic patterns, spatially-dependent climatic factors not
 407 captured by the predictors, or even systematic biases in the inventory mapping. The higher
 408 significance of the LM-error test indicates that a model addressing spatially autocorrelated errors
 409 (e.g., SEM) may be more appropriate than one modeling endogenous interaction (e.g., SLy).

410 **4.4 Spatial regression models**

411 We evaluated landslide susceptibility regression models using two main approaches: (1) Spatial
 412 heterogeneity models, which allow regression coefficients to vary across space using spatial regime
 413 models (SR) and Geographically Weighted Regression (GWR) models, and (2) Spatial dependence
 414 models, which account for spatial interactions between neighboring units using spatial
 415 autoregressive models (SLy, SEM, SAC, SLX, SDEM, SDM, GNS). Finally, we combined both
 416 approaches, integrating spatial dependence and heterogeneity into a single framework (Table 2).

417 Table 2: Summary of the models used in the study and the figures/tables presenting their results.

Model	Description	Results in
(non-spatial)	Standard regression without spatial effects. Baseline model for comparison.	Table 1, Fig. 8
Spatial heterogeneity models		
Spatial Regime (SR)	Models with intercepts varying by regions (basins/clusters).	Fig. 9
	Models with intercepts and slopes varying by regions (basins/clusters).	Fig. 10

Model	Description	Results in
Geographically Weighted Regression (GWR)	Models with coefficients varying locally using several bandwidths (BW = 10, 50, 200, 500).	Fig. 11
	Models with coefficients varying locally using different bandwidths (Area BW=124, Elevation BW=302, Relief BW=504).	Fig. 12
Spatial dependence models		
SLy	Model with spatial autocorrelation in the dependent variable.	Table 3
SEM	Model with spatial autocorrelation in the errors.	Table 3, Fig. 14
SLX	Model with spatial autocorrelation in the predictors.	Table 3
SDEM	Spatial Durbin error model, combining SEM with spatial effects on predictors.	Table 3
SDM	Spatial Durbin model, combining Sly with spatial effects on predictors.	Table 3
SAC	Combined model with spatial dependence in both the dependent variable and error term.	Table 3, Table 4
GNS	Model combining spatial effects in the dependent variable, the errors and the predictors.	Table 3
Combined model	Integration of SR using clusters and autoregressive models (SAC, SEM, SDEM).	Fig. 13

4.4.1 Regression models for spatial heterogeneity

We implemented spatial fixed-effect (intercept-only) models independently for the basin-based regimes and the morphometric cluster-based regimes (Figure 9). This allowed us to separately evaluate how each spatial partitioning scheme accounts for regional variations in landslide occurrence. In the basin regime model, the effects for the Cauca and Magdalena basins have comparable magnitudes, suggesting similar regional influences on landslide abundance. In the cluster regime model, the spatial regimes for clusters B and D show stronger marginal effects than those of morphometric predictors such as area (A), mean elevation (E), and mean relief (H), indicating grouping catchments by these properties captures important spatial variability (Figure 10). For example, H has a much higher coefficient in the Magdalena and Cauca basins, while E is higher for the Atrato and Cauca basins. For the cluster regimes, E is not statistically significant in any cluster. However, the SR model based on cluster regimes has a slightly better performance. The SRs model for cluster regimes slightly raises adjusted R^2 and reduces AIC, with only a marginal difference to the best-performing OLS model.

A key result from the Geographically Weighted Regression (GWR) is a surface of local parameter estimates for each predictor (Figure 11). we observe a clear trade-off: smaller bandwidths (e.g., $BW=10$) yield a better statistical fit (higher Adj. R^2 , lower AIC) but produce highly uncertain parameter estimates. Conversely, larger bandwidths (e.g., $BW=500$) provide more stable coefficients, though at the cost of a poorer model fit. These results indicate that lower bandwidths in GWR increase multicollinearity among predictor variables; as the estimation window shrinks, predictors often exhibit higher redundancy within the local subset than they do across the entire dataset. This leads to unstable parameter estimates, counterintuitive parameter signs, overconfident R^2 values, and inflated standard errors (Kyriazos & Poga, 2023).

443 Figure 12 shows the spatial variation in the coefficients of **A**, **E** and **H** for different bandwidths,
444 and reveal spatial non-stationarity. We observe a positive relationship between **A** and landslide
445 abundance throughout most of the region. The influence of **E** is stronger in the northeastern and
446 southwestern parts of the study area, whereas the effect of **H** increases toward the northwest and
447 decreases toward the southeast.

448 This observed spatial non-stationarity provides insights into the interplay of tectonic, climatic, and
449 geomorphological processes that are typically masked by the averaged estimates of global models.
450 The varying coefficient surfaces (Fig. 12) suggest that the drivers of landsliding are not uniform.
451 For example, the increasing influence of **H** towards the northwest corresponds to the highly
452 dissected, tectonically active terrain of the Cauca canyon (Cluster B). In contrast, the influence of **E**
453 is strongest in the northeastern and southwestern high-altitude plateaus (Cluster D). A single, global
454 coefficient for **E** or **H** (Table 1) is likely too simplistic. Landslide susceptibility in this region is
455 controlled by different dominant processes in different geomorphological domains—a finding
456 achievable through heterogeneity models like GWR or SR.

457 **4.4.2 Regression models for spatial dependency**

458 Table 3 summarizes the estimates of the spatial autoregressive (SAR) models. While the spatial
459 autoregressive error models (SEM and SDEM) yield the lowest AIC values, the improvement in
460 model fit compared to the other spatial models is modest. Despite accounting for spatial
461 autocorrelation in these models, the overall difference among them is negligible.

462 The spatial lag of exogenous predictors refers to including neighboring values of the predictor
463 variables, thus capturing potential spillover effects from surrounding areas. In theory, this allows the
464 model to account for the influence that environmental factors in adjacent catchments may have on
465 landslide abundance at a given location. For example, high rainfall accumulation in one catchment
466 could affect slope saturation in neighboring catchments, indirectly contributing to landslide

467 occurrence. We find that the coefficients for the spatially lagged predictors are statistically
 468 insignificant, indicating that the influence of predictors is primarily local rather than being
 469 influenced by those in neighboring areas.

470 Table 3: Model Results for SAR Models

Predictors	SLy	SEM	SAC	SLX	SDEM	SDM	GNS
Constant	0.824 (0.08)	1.819 (0.11)	1.658 (0.23)	1.841 (0.05)	1.812 (0.11)	0.687 (0.09)	2.342 (0.31)
A	0.530 (0.04)	0.552 (0.04)	0.558 (0.04)	0.551 (0.05)	0.561 (0.04)	0.557 (0.04)	0.560 (0.04)
E	0.263 (0.05)	0.539 (0.09)	0.502 (0.10)	0.440 (0.19)	0.391 (0.15)	0.409 (0.16)	0.381 (0.15)
H	0.281 (0.05)	0.469 (0.08)	0.457 (0.08)	0.446 (0.12)	0.395 (0.09)	0.388 (0.10)	0.410 (0.09)
W _X -A				0.047 (0.12)	0.020 (0.12)	-0.317 (0.10)	0.185 (0.15)
W _X -E				0.076 (0.21)	0.167 (0.20)	-0.192 (0.18)	0.345 (0.23)
W _X -H				0.128 (0.14)	0.204 (0.14)	-0.157 (0.12)	0.368 (0.18)
ρ	0.561		0.089			0.621	-0.295
λ		0.621	0.565		0.621		0.751
Nagelkerke R ²	0.62	0.628	0.628		0.631	0.631	0.633
AIC	1468.111	1456.317	1457.932	1615.105	1458.16	1458.759	1457.047
<i>p</i> <0.05, (standard errors)							

471
 472 Finally, we combined spatial dependency and spatial heterogeneity by integrating the spatial lag
 473 model (SAC), spatial error model (SEM), and Spatial Durbin Error Model (SDEM) with the spatial
 474 regime (SR) model using clusters as regimes (Figure 13). We selected the cluster-based regime for
 475 this combination because it outperformed the basin-based regime in other models (Figure 10),
 476 capturing more relevant morphometric variability. Yet, the combined model provides only marginal
 477 improvements over the standalone SAR models. The SAR models already explains much of the
 478 spatial structure in the data, limiting the additional benefit from explicitly modeling spatial
 479 heterogeneity through the cluster regimes. Although the cluster regime adds information about
 480 localized variations in morphometric conditions, the gains in predictive performance remain

modest, indicating that spatial dependency explains more of the landslide occurrence in the study area.

Figure 14 shows the landslide abundance estimated by the SEM model and the distribution of residuals, which have no spatial structure ($MI = -0.015$ with $p = 0.71$).

Table 4 presents the global spillover effect for spatial autoregressive models with endogenous interaction effects of the response variable (SLy, SAC, SDM and GNS). The spillover effects must be interpreted with caution, avoiding a purely statistical reading. The indirect effects (spillovers) reported here do not imply that, for example, high relief in catchment i physically cause landslides in catchment j . Rather, these terms estimate the impact of spatially autocorrelated predictors. The positive indirect effects for E (0.152) and H (0.206) mean that catchments with high relief tend to be geographically clustered with other catchments that also have high relief (as supported by Moran's I for these predictors). The model is thus indicating that the landslide susceptibility in a given catchment is influenced by both its own attributes (direct effect) and those of its surrounding geomorphological neighborhood (indirect effect).

Table 4: Global spillover effect

Predictors	Direct	Indirect	Total
Mean Area (A)	0.561	0.070	0.631
Mean elevation (E)	0.419	0.152	0.572
mean relief (R)	0.401	0.206	0.608

5. Discussion

5.1 Limitations of the non-spatial framework

We used a standard linear regression as a baseline to examine the relationship between landslide abundance and several predictors. Although these models feature elevation, relief, and catchment area as key predictors, their overall explanatory power is limited (Table 1), likely due to the non-Gaussian distribution of the response variable.

One point for discussion is the apparent statistical insignificance of geology (G) and land use (Lc) in the parsimonious models, a finding that may seem physically counter-intuitive. We argue this result should not be interpreted as a lack of geomorphological relevance, but rather as a demonstration of scale-dependent effects in susceptibility modeling. At the slope-scale, geological and land-cover variables are often dominant. However, at the catchment-scale of this study (median area: 48 km²), their explanatory power appears to be overwhelmed by the dominant, macro-scale morphometric gradients (E and H). This scale-dependent effect is likely attributable to two factors. First, using the modal class for Lc and G within a large catchment unit neglects the variety of classes. Second, at the regional scale, processes related to tectonic uplift and climate, which are expressed through macro-scale topography (E and H), appear as first-order controls on landslide abundance. G and Lc may therefore represent second-order, local controls that are inadequately captured by this aggregated modeling approach, rather than irrelevant factors.

Finally, the OLS residuals show a clear, non-random spatial structure, with a significant global Moran's I of 0.376 ($p = 0.001$). This residual autocorrelation shows that the OLS model is misspecified, as it fails to capture local variations and violates the assumption of independent errors. This pattern is not uniform; exploratory analysis (Fig. 8) shows significant overestimation in the high-relief, complex terrain of clusters B and D .

5.2 Interpreting spatial structure

We interpret the two forms of spatial structure that our models addressed: heterogeneity and dependence. This is not merely a statistical exercise; it is a diagnostic effort to understand *why* the OLS model failed and what physical processes it obscured.

5.2.1 Interpreting spatial heterogeneity (GWR and SR models)

Model results confirm that the influence of geomorphological drivers is not uniform across the study area. By allowing coefficients to vary, GWR (Figs. 11, 12) and SR models (Figs. 9, 10) revealed distinct geomorphological domains where different processes are dominant.

For instance, the explanatory power of mean relief (**H**) is highest in the highly-incised, tectonically active terrain of the Cauca canyon (Cluster B), while mean Elevation (**E**) exerts a primary influence in the plateaus of the Central Cordillera (Cluster D). This demonstrates that a single, global coefficient for these predictors (as in Table 1) is a geomorphological oversimplification, averaging out the distinct processes that control landslide occurrence in different regions. Fotheringham et al. (2009) refer to this type of variability as different levels of spatial non-stationarity, where some predictors exhibit stronger local variations over short distances, while others may appear more stable.

While GWR offers a continuous surface of this variability, its results can be sensitive to bandwidth selection; narrow bandwidths risk parameter instability and multicollinearity, while wide bandwidths over-smooth local effects (Chen et al., 2024). While recognizing the potential for local instability in GWR estimates, these bandwidth variations suggest potential differences in the spatial scales at which each variable conditions landslide susceptibility. The relatively narrow bandwidth for variable **A** suggests that its influence might be more localized, potentially aligning with the spacing of catchment divides, whereas the moderate bandwidth for elevation may reflect broader regional topographic trends and their interaction with rainfall. In contrast, the larger bandwidth for mean relief (**H**) suggests a more consistent effect across broader areas, consistent with the smoother spatial variation of macro-scale morphometry compared to localized features. However, these scale-dependent interpretations remain exploratory, as the narrower bandwidths are also where the model is most sensitive to local collinearity. The SR model, while dependent on *a priori* region definition (e.g., our geomorphological clusters), avoids this instability and offers a clearer, more interpretable comparison of effects between these distinct process domains.

550 5.2.2 Interpreting spatial dependence (SAR models)

551 The spatial dependence models address the assumption of independent observations. Our analysis,
552 guided by the LM tests (Section 4.3), found that the Spatial Error Model (SEM) had the most
553 parsimonious fit (lowest AIC) among the tested SAR variants. A superior fit for a spatial lag model
554 (e.g., SLy) would have implied a 'contagion' process, where high landslide frequency in one
555 catchment directly influences the frequency in its neighbor. While a direct physical contagion for
556 landslides—where one event triggers another across a catchment divide—is rare, this model can be
557 appropriate if theory suggests a spatially contagious process is acting as a proxy for unobserved,
558 clustered phenomena like regional deforestation patterns or cascading hillslope adjustments. The
559 better performance of the SEM suggests this is not the dominant process, pointing instead to
560 autocorrelated drivers as the most physically justifiable explanation.

561 The significant λ term in the SEM is effectively a proxy for latent, spatially-clustered drivers that
562 were not included in our model. These could include unmeasured factors such as regional jointing
563 patterns, localized storm cells not captured by the 5-km CHIRPS data, or systematic, spatially-
564 clustered biases in the landslide inventory itself. Thus, the SEM not only corrects the model but also
565 serves as a diagnostic tool to highlight the influence of these latent factors.

566 5.3 Practical implications for landslide susceptibility

567 The findings of this study have direct and practical implications for regional landslide susceptibility
568 assessment, moving beyond statistical metrics to inform resource allocation and mitigation
569 planning. First, our results challenge the validity of applying global susceptibility models to large,
570 complex regions like the Andes. A non-spatial OLS model, for example, would incorrectly assign a
571 single, uniform weight to relief for the entire study area. A regional susceptibility manager using
572 this global model might misallocate resources by focusing on high-relief areas in regions where, as
573 our GWR and SR results demonstrate (Figs. 10, 12), Mean Elevation (E) is the true dominant factor

574 (e.g., Cluster D). Second, the strong evidence for spatial heterogeneity (non-stationarity) provided
575 by the GWR and SR models suggests that a global mitigation strategy is inherently flawed. Instead,
576 susceptibility mapping and subsequent land-use planning should be stratified by the distinct
577 geomorphological domains identified in our clustering analysis (Fig. 7). Mitigation efforts in the
578 highly-incised Cauca canyon (Cluster B) should prioritize factors related to high local relief and
579 associated slope processes, while efforts in the Central Cordillera (Cluster D) may need to focus on
580 processes occurring at high elevations, which may be linked to different rainfall patterns or bedrock
581 properties.

582 Finally, the better performance of the Spatial Error Model (SEM) serves as a diagnostic tool for
583 practitioners. It provides a quantitative warning that the selected, measured predictors (A, E, H),
584 while significant, are insufficient to fully explain the observed landslide patterns. This finding
585 highlights the need to invest in acquiring better, more granular data on the latent drivers the SEM is
586 capturing. This could include, for example, high-resolution rainfall data (to capture localized
587 storms), detailed lithological or structural maps (to capture regional joint patterns), or improved
588 inventory mapping protocols to reduce systematic spatial bias.

589 **5.4 Methodological considerations and future research**

590 While our findings demonstrate the value of integrating spatial structure, several methodological
591 considerations and avenues for future research must be addressed.

592 **(i)** The quality and potential biases of the landslide inventory are a primary concern. Any spatial
593 model will amplify, not correct, systematic spatial biases present in the data, such as a concentration
594 of mapped landslides along infrastructure (Caleca et al., 2025). While spatial models can be
595 strategically used to isolate this bias, our study proceeds with the assumption that the inventory,
596 while incomplete, is a spatially representative sample. Future work should explicitly test the
597 sensitivity of these models to known inventory biases.

598 **(ii)** The definition of the spatial weights matrix (W) is a subjective step. Our data-driven choice of a
599 $k=5$ nearest-neighbor matrix was based on maximizing spatial autocorrelation. However, this choice
600 is not neutral; it pre-defines the scale and nature of spatial interactions. Future research could
601 explore physically-grounded matrices based on, for example, hydrological connectivity or
602 geomorphological similarity, to test specific process-based hypotheses.

603 **(iii)** Our justification for selecting the SEM is further supported by a robust statistical model
604 selection process. When we tested combined models (e.g., SAC) that include terms for both error
605 (λ) and endogenous (ρ) dependence, we found that the spatial lag coefficient (ρ) decreased
606 significantly while the spatial error coefficient (λ) remained stable and significant. This strongly
607 indicates that spatial dependence in our data is primarily captured by the error term, reinforcing our
608 conclusion that the process is driven by latent, unmeasured spatial factors rather than a true
609 contagion effect.

610 **(iv)** While we explored models for heterogeneity (SR) and dependence (SEM) separately, we also
611 tested combined frameworks (e.g., SR-SEM)⁸. This approach, which separates large-scale
612 heterogeneity from fine-scale dependence, introduces significant complexity, multicollinearity, and
613 overfitting. In our case, the gains in performance were marginal. However, future work should
614 explore hybrid methods that can more adaptively model these dual spatial effects, especially in
615 highly complex terrains.

616 **(v)** This study was conducted within a Gaussian-based regression framework, requiring a
617 $\log(Y+1)$ transformation of the discrete, zero-inflated landslide count data. We acknowledge that
618 this transformation is an approximation and does not fully account for the underlying data-
619 generating process. This approach was adopted due to the practical availability of robust spatial
620 econometric software. A critical avenue for future research is the application of more sophisticated
621 spatial regression frameworks explicitly designed for count data, such as Spatial Poisson, Negative
622 Binomial (NB), or Zero-Inflated (ZINB) models. Such models would provide a more statistically

623 rigorous estimation of landslide frequency, eliminating the need for data transformation and offering
624 deeper insights into the complex spatial patterns of landslide susceptibility.

625 **(vi)** Finally, while our analysis identified the SEM as the most parsimonious model, other variants
626 like the Spatial Durbin Model (SDM) or Durbin Error Model (SDEM) remain highly relevant for
627 future research. These models, which account for exogenous spillovers, are ideal when observation
628 suggests strong spatial interactions between predictors. For example, they could explicitly model
629 processes such as hydrological connectivity (where high rainfall in an upstream catchment affects a
630 downstream one) or infrastructure interdependence (where a road network propagates risk).
631 Exploring these models could offer even deeper insights where such cross-catchment processes are
632 dominant.

633 **6. Conclusion**

634 Our analysis demonstrates that the spatial structure of landslide data, often ignored in susceptibility
635 modeling, is not a statistical nuisance but a fundamental component of the geomorphological
636 system. We have shown that standard, non-spatial models are fundamentally misspecified, as
637 evidenced by significant residual autocorrelation and their failure to capture scale-dependent
638 effects.

639 By integrating spatial models, we move beyond a single, global model to reveal that landslide
640 controls are spatially heterogeneous, with relief dominating in high-incision terrains and elevation
641 controlling susceptibility on high-altitude plateaus. Furthermore, the dominance of spatial error
642 (SEM) provides a powerful diagnostic, suggesting that landslide patterns are strongly driven by
643 latent, spatially-clustered factors, such as tectonic fabrics, localized storm cells, or inventory biases
644 We conclude that explicitly accounting for both spatial heterogeneity and dependence is a necessary
645 step to advance landslide susceptibility from a statistical pattern-matching exercise to a more
646 realistic, process-based susceptibility assessment. Future work must build on this by integrating

647 non-Gaussian frameworks and by using spatial models to strategically investigate the impact of
648 inventory quality on susceptibility assessment.

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652 **Competing Interests**

653 The authors declare no competing interests.

654 **Author Contributions**

655 EA designed the study, compiled the landslide inventory, and led the spatial modeling and statistical
656 analysis. LL and OK contributed to the conceptual framework of spatial models and supervised the
657 implementation of spatial regression techniques. OK revised the interpretation of results, and
658 supported the writing and structuring of the discussion. All authors contributed to the manuscript
659 preparation, reviewed, and approved the final version.

660 **Open Research**

661 Data and code used for this study are fully available in Github
662 (https://github.com/edieraristizabal/PAPER_SAR).

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Fig. 1 Overview of spatial regression models that incorporate spatial heterogeneity (pink) and spatial dependency (blue). Spatially-lagged Conditional Autoregressive Models (CAR) or Simultaneous Autoregressive models (SAR) target specific spatial structures: CAR models include options like Intrinsic CAR, Besag-York-Mollie (BYM), and Leroux, while SAR models account for spatial lag in spatial connectivity or network structures; by adding terms for spatial autocorrelation in the response variable WX , the local error $\lambda W\mu$, or spatial lag effects ρWy , we obtain different variants of spatial regression models.

Fig. 2 Location of the study area and 13,777 landslides between 1970 and 2023. The first map (A) illustrates the spatial distribution of individual landslides across the study region. The inset shows the geographical location of the study area within South America. Map B shows the abundance of landslides aggregated by catchments.

Fig. 3 Histogram of landslide abundance per catchment (A), highlighting high skewness with most observations clustered near zero, i.e. no detectable landslides in a given catchment. The panel B shows transformed variable, i.e. $\log(\text{Landslide abundance} + 1)$; black line is a Gaussian kernel density estimate.

Fig. 4 Principal Component Analysis (PCA) biplot of catchment characteristics (the hypsometric integral (H_i), mean elevation (E), mean slope S , mean local relief (H), lineament density (L_d), mean annual rainfall (P), land use cover (L_c), geology (G), the number of days with accumulated daily rainfall exceeding 20 mm (P_{20}). The first two principal components (PC1 and PC2), explain 49% and 21% of the total variance. Each bubble represents a catchment, with red circles indicating catchments with reported landslides ($\text{Landslides} > 0$) and green circles indicating those without ($\text{Landslides} = 0$). Bubble size is scaled to the number of landslides in each catchment; arrows show the loading of catchment variables on the principal components.

Fig. 5 Five-nearest-neighbour ($k = 5$) connectivity representing the neighboring relationships between catchments in the study area (blue nodes = centroids, red edges = neighbour links). Spatial connectivity network.

Fig. 6 Local Indicator of Spatial Autocorrelation (LISA) and the Moran Index for the slope angle. Red color means high values surrounded by high values (Q1-hotspot), blue color means low values surrounded by low values (Q3-coldspot), blue and yellow color mean high values-low values (Q2) and low values-high values (Q4), respectively. Gray dots indicate locations (catchments) where the spatial autocorrelation is statistically insignificant.

Fig. 7 Spatial regimes used in the study area: (A) basin boundaries and (B) morphometric clusters of 533 catchments. The left map shows the delineated boundaries of the major river basins in the study area, including the Atrato, Cauca, and Magdalena basins, represented by green, red, and blue, respectively. These basins constitute the first spatial regime. The right map displays the second spatial regime, derived from a cluster analysis of morphometric properties using a K-Nearest Neighbors (KNN) algorithm ($K = 5$), which identified four clusters (A, B, C, and D) that group catchments with similar morphometric characteristics.

Fig. 8 Non-spatial OLS model standardized residuals for basins and clusters. The top panels (A, B) show the spatial distribution of model residuals across the study area. Positive residuals (in blue) indicate overestimation by the model, while negative residuals (in red) indicate underestimation. The polygons' outline in the map is color coded according to the three basins. The panels C and D show box plots of standardized residuals across the Atrato, Cauca, and Magdalena basins: non-overlapping notches identify differences in median values.

Fig. 9 Spatial fixed effects with estimated standard errors for the cluster and basin-regime models. The panel A displays results for the cluster regimes, and the panel B shows the basin regimes.

Fig. 10 Group-level intercepts and slope models for spatial regimes (Basin and Cluster), showing parameter estimates with standard errors.

Fig. 11 Parameter estimates for a Geographically Weighted Regression (intercept, Area (A), Elevation (E), Relief (H) with varying bandwidths in terms of number of catchments (BW = 10, BW = 50, BW = 200, BW = 500). Points are the means, and bars indicate the standard error for each predictor within each bandwidth.

Fig. 12 Spatial variation of Multiscale Geographically Weighted Regression coefficients for (A) catchment area (BW = 124), (B) mean elevation (BW = 302), and (C) mean relief (BW = 504). BW in terms of neighbor catchments for each predictor were iteratively optimized in an adaptive process.

Fig. 13 Parameter estimates for SEM, SLy, and SDM models. Bubbles are mean estimates, bars are their standard errors. Other model-specific parameters are: λ for SEM (blue); ρ for SLy (green); and SDM (red), along with Nagelkerke R² values for SLy and SDM models, and AIC values for each model. WX corresponds to the spatially lagged predictors, which captures the exogenous interaction effects among the predictors.

Fig. 14 Maps (A) of SEM residuals color-coded to the deviations between predicted and observed landslide occurrences. Negative residuals (blue) show underprediction, while positive residuals (yellow) show overprediction. Map (B) of the expected landslide abundance.





























